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Hot-Stove Effect, Sampling Asymmetry, and Feedback in Reinforcement Learning

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Background: what the hot-stove effect is in reinforcement learning and decision making

The hot-stove effect is a learning trap: one bad early outcome can make a person or agent avoid an option, which then blocks the later evidence that might have shown the option was actually good on average. In reinforcement learning and repeated choice, it is a self-reinforcing error caused by learning from what is chosen while learning little or nothing from what is avoided. (6 sources)

In reinforcement learning and decision making, the hot-stove effect refers to a pattern where a strongly negative early experience makes an option seem worse than it really is, so the decision maker stops choosing it and never gathers the evidence needed to correct that belief. The classic intuition is Mark Twain's cat: after sitting on a hot stove once, it never sits on one again, and thus never learns that most stoves are harmless most of the time. Denrell and later work frame this as an asymmetry that appears when sampling is costly: if an option looks bad early, it is dropped and under-sampled; if it looks good early, it continues to be sampled and its estimate can eventually be corrected. (Wright et al., 2017) (Hertwig et al., 2021)

In simple terms, the bias comes from avoidance after bad outcomes. A bad draw lowers the option's perceived value, which reduces future choice of that option; once it is no longer chosen, little or no new information arrives, so the bad impression remains both recent and salient. This can create a self-reinforcing loop of undersampling, misvaluation, and further undersampling. (Weiss-Cohen et al., 2020) (Govoni et al., 2026)

In bandit-like or explore–exploit settings, the effect is most often discussed for risky options that have higher variance: a few low payoffs from the risky arm can push the learner into the safer arm, where they then get "stuck." Toyokawa et al. describe this as a fundamental consequence of adaptive sampling: once decision makers underestimate the risky option, they avoid it and lose the chance to fix the error. (Toyokawa et al., 2021)

More broadly, the hot-stove effect is one way to describe how learning from experience can produce persistent avoidance that looks like overweighting a rare bad event, even when the event is not common. How strong the effect becomes depends on the structure of the task, including whether the learner can observe what would have happened under the unchosen action. (Hertwig et al., 2021) (Godara, 2025)

Evidence

(Wright et al., 2017)

Go for broke: The role of somatic states when asked to lose in the Iowa Gambling Task.

"(Denrell, 2005) identifies an intriguing asymmetry that arises when sampling information is costly (as in the IGT, where turning a card to learn more about a deck risks forgoing a better option). If an option initially appears good, people will keep selecting it, though if initial outcomes are better than the longrun average for that option they will eventually learn to disfavor that option. In contrast, if an option initially appears bad, people will stop selecting it, and so forgo the opportunity to learn that the option is, on average, better than they initially experienced. This is termed the "hot stove effect" (Denrell & March 2001), following an anecdote from Mark Twain who observed that a cat need only sit once on a hot stove for it to never to do so again, but it will consequently never learn that most of the time the stove is a perfectly decent place to rest."

(Hertwig et al., 2021)

A Description–Experience Framework of the Psychology of Risk

"The hot-stove effect refers to a behavior that initially yielded an extremely adverse outcome and can therefore give rise to a powerful behavioral bias that prevents the organism from repeating this behavior (Denrell, 2005)(Denrell, 2007)(Denrell et al., 2001). A cat that sits on a hot stove lid may never approach another one, regardless of its temperature (Twain, 1897, p. 124). The cat behaves as if it overweights the (possibly) rare event of getting burned. The likelihood of a hot-stove effect depends on a number of factors, such as whether the organism can take precautionary actions (e.g., turning off the stove), the causal model the organism uses (e.g., believing that lightning never strikes twice), and the kind of feedback they receive (e.g., whether they get feedback on what would have happened had they chosen the forgone action)."

(Weiss-Cohen et al., 2020)

Timing of descriptions shapes experience-based risky choice

"Another behavioral phenomenon which can interfere in the learning process is the hot stove effect (Denrell et al., 2001)). 2 The hot stove effect states that bad outcomes reduce the attractiveness of an option, causing those options to be avoided (Biele et al., 2009). By avoiding an option after a bad outcome has occurred, the ability to collect more information about that option is severely constrained, as feedback from that option will no longer be observed. With no new information beeing collected, the bad outcome observed will remain the most recent (and most salient) observation associated with that alternative, decreasing its perceived value"

". The hot stove effect severely constrains the learning process by reducing the likelihood of selections from undesirable options, after a sequence of bad outcomes; in partial-feedback environments, there are no opportunities for further learning from unselected options, therefore eliminating learning about how unnattractive these options really are (i.e., how often the stove is hot; (Denrell et al., 2001). This process is likely to be related to the overweighting of rare events observed in decisions from descriptions, which would dictate that even though the described bad outcomes only occur rarely, individuals perceive them as happening much more often (4635002). In contrast, in paradigms, the hot stove effect is not sustained because individuals are able to learn from the foregone outcomes and therefore realize that the stove is not necessarily "hot" all the time. Eliminating the hot stove effect via full-feedback in our experiment was detrimental to performance, leading to more risk-taking, lower bonuses, and reduced impact of the descriptions, which acted as warning messages."

(Govoni et al., 2026)

From Optimal Policies to Individual Differences: Rethinking Reinforcement Learning for Biology

"A classic illustration is the "hot stove effect" (Denrell et al., 2001), i.e., the tendency to neglect an option after an initial bad outcome. Touching a hot stove can give rise to a self-reinforcing loop: undersampling leads to incorrect valuation ("all stoves are hot"), driving further undersampling ("I will never come close to a stove again")."

(Toyokawa et al., 2021)

Conformist social learning leads to self-organised prevention against adverse bias in risky decision making

"Under such circumstances, the hot stove effect occurs: Experiences of low-value payoffs from the risky option tend to discourage decision makers from further choosing the risky option, trapping them in the safe alternative"

". emergence of the risk-seeking equilibrium, defined as □ < □ ¸could become harder with decreasing exploration rate , consistent with the known property of asymmetric adaptive sampling that causes the hot stove effect (Denrell, 2007)"

". Overall, our model crystallises the asymmetry emerging from adaptive sampling, which is considered as a fundamental mechanism of the hot stove effect (Denrell, 2007;March, 1996): Once decision makers underestimate the expected value of the risky alternative, they start avoiding it and do not have another chance to correct the error"

". Information about others' payoffs might also be available in addition to inadvertent social frequency cues in some cooperative contexts, especially with the aid of benevolent pedagogical instructions from others. Although communicative acts may transfer information about unexplored options and forgone payoffs, which potentially mitigates the hot stove effect (Yechiam and Busemeyer, 2006)"

(Godara, 2025)

Apparent learning biases emerge from optimal inference: Insights from master equation analysis

"This task captures the explore-exploit dilemma faced by all decision-makers operating in uncertain environments"

". In particular for tasks without counterfactual information, we have u ± = 0."

Mechanisms: ordinary sampling asymmetry versus asymmetric loss or gain treatment

The hot-stove effect does not have to mean that people intrinsically weight losses more than gains. A lot of it can arise from plain sampling asymmetry: chosen options get learned about, avoided options do not, and this alone can create persistent avoidance, though some tasks may also show extra asymmetries in how positive and negative outcomes are updated. (4 sources)

A key point is that the hot-stove pattern can come from the structure of experience itself, not only from a built-in tendency to treat bad outcomes as more important than good ones. When one option is sampled less, its value estimate stays more uncertain, and that uncertainty can itself push people away from it, which is exactly the kind of self-reinforcing under-sampling highlighted by hot-stove accounts. Aberg et al. note that it is still unclear whether this effect in simple RL tasks arises from ordinary differences in sampling rate alone or instead needs “special circumstances” that create option-dependent weighting of gains and losses; they also note that prior work has suggested the latter may sometimes be needed. (Aberg et al., 2025)

At the same time, there is strong evidence that asymmetric updating can amplify these dynamics when it is present. In RL terms, if learning rates are larger for outcomes that support the current choice and smaller for outcomes that would weaken it, then choices become self-confirming: good outcomes for chosen options and bad outcomes for unchosen options get more weight, which promotes choice repetition. Sugawara et al. summarize this as a confirmation-like asymmetry, and Katahira shows more generally that differential learning rates induce autocorrelation in choice, meaning repeated choosing or switching beyond what symmetric updating would predict. (Sugawara et al., 2021) (Katahira, 2018)

So the clean synthesis is: ordinary sampling asymmetry is enough to generate a hot-stove-style trap, because avoidance cuts off corrective evidence; asymmetric treatment of gains and losses is not conceptually required, but it can strengthen the trap by making belief updates more choice-confirming. This also helps explain why the “best” asymmetry depends on the feedback regime. Lefebvre et al. report that with full feedback about both chosen and unchosen options, reward is often maximized by a confirmation-bias pattern of learning rates, whereas in standard bandit feedback tasks the useful direction of asymmetry depends on the reward probabilities, with optimistic or pessimistic bias helping in different environments. (Lefebvre et al., 2020)

Evidence

(Aberg et al., 2025)

Estimation-uncertainty affects decisions with and without learning opportunities

"Therefore, upon observing a tendency to choose one option over another, one cannot a priori exclude that estimations of expected outcomes contributed only partly to this tendency. Another contributing factor may have been uncertainty associated with the expected outcome estimations. Specifically, all else being equal, if an option has been sampled less than another then uncertainty related to its outcome estimation is increased, and so the less-sampled option might be chosen less often. This possibility is consistent with the 'hot-stove effect' theory in reinforcement learning, which states that infrequently sampled options are treated as potentially worse than options sampled more frequently, and so are avoided36. However, whether the hot stove effect in simple reinforcement learning tasks emerges due to ordinary sampling-rate asymmetries between options (e.g., due to one option being of higher expected value than the other), or requires special circumstances that instigate option-dependent weighting of gains and losses, is unclear. Previous studies have suggested that the latter condition may be required23,36,37.

Unlike ordinary RL, however, in standard binary choice tasks with full and immediate feedback for both options after each choice, sampling rates during learning are fully under experimental control and are frequently held equal across options. So, the role of this source of uncertainty in such tasks is generally unclear"

"Two prior studies have investigated this issue. One of these did not identify a clear impact of differences in uncertainty13, whereas the other found that uncertainty does influence choices following learning, but employed a task that included feedback about outcomes both of chosen options and of unchosen options38. It therefore remains to be shown whether uncertainty can affect long-term choices following ordinary trial-and-error reinforcement learning, where feedback about outcomes is available only for chosen actions ('bandit feedback')."

(Sugawara et al., 2021)

Dissociation between asymmetric value updating and perseverance in human reinforcement learning

"These learning asymmetries lead to choice repetition because the influences of the outcomes that reinforce the choice (positive outcome for the chosen option and negative outcome for unchosen option) are enhanced, whereas those that weaken the choice are diminished (Katahira, 2018) ."

(Katahira, 2018)

The statistical structures of reinforcement learning with asymmetric value updates

"Abstract Reinforcement learning (RL) models have been broadly used in modeling the choice behavior of humans and other animals. In standard RL models, the action values are assumed to be updated according to the reward prediction error (RPE), i.e., the difference between the obtained reward and the expected reward. Numerous studies have noted that the magnitude of the update is biased depending on the sign of the RPE. The bias is represented in RL models by differential learning rates for positive and negative RPEs. However, which aspect of behavioral data that the estimated differential learning rates reflect is not well understood. In this study, we investigate how the differential learning rates influence the statistical properties of choice behavior (i.e., the relation between past experiences and the current choice) based on theoretical considerations and numerical simulations. We clarify that when the learning rates differ, the impact of a past outcome depends on the subsequent outcomes, in contrast to standard RL models with symmetric value updates. Based on the results, we propose a model-neutral statistical test to validate the hypothesis that value updates are asymmetric. The asymmetry in the value updates induces the autocorrelation of choice (i.e., the tendency to repeat the same choice or to switch the choice irrespective of past rewards). Conversely, if an RL model without an intrinsic autocorrelation factor is fitted to data that possess an intrinsic autocorrelation, a statistical bias to overestimate the difference in learning rates arises. We demonstrate that this bias can cause a statistical artifact in RL-model fitting leading to a "pseudo-positivity bias" and a "pseudo-confirmation bias.""

(Lefebvre et al., 2020)

A normative account of confirmation bias during reinforcement learning

"In this paper we show that if the participants are able to view the payouts associated with both chosen and unchosen options, reward is typically maximized if the learning rates follow the pattern of the confirmation bias, i.e. are higher when the chosen option is rewarded and unchosen option is unrewarded"

".Caze and van der Meer (2013) considered a standard reinforcement learning task in which feedback is provided only for the chosen option. In that task the reward probabilities of the two options in the task determine whether it is beneficial to have higher learning rate after positive or negative prediction error (Cazé et al., 2013). In other words, when only outcome of chosen option is observed, optimistic bias is beneficial for some reward probabilities, while pessimistic bias for other."

Feedback regimes: bandit or partial feedback versus full or counterfactual feedback in preventing self-reinforcing sampling errors

Hot-stove errors are hardest to escape under bandit or partial feedback, because the learner only sees outcomes for chosen actions and gets no corrective evidence about avoided ones. Full or counterfactual feedback can break this loop by revealing foregone outcomes, though that does not always improve reward because it can also remove useful caution and increase risk taking. (7 sources)

The clearest divide is between **bandit/partial feedback** and **full/counterfactual feedback**. In bandit settings, the learner observes only the reward and next state for the action actually taken, not what the other action would have produced, whereas full-feedback settings reveal outcomes more broadly and are known to be much easier from a learning standpoint than bandit feedback. (Lykouris et al., 2019) This matters directly for the hot-stove effect: if an option is avoided after an early bad result, then under partial feedback there is no new evidence from that option to correct the belief, so the bad impression can persist and keep driving avoidance. (Weiss-Cohen et al., 2020) (Aberg et al., 2025)

By contrast, **full or counterfactual feedback can prevent the self-reinforcing sampling error itself**, because the learner can observe foregone outcomes from unchosen options without having to sample them. That

removes the key informational asymmetry behind the trap: the option may be avoided in choice, but it is no longer unseen in learning. Experiments summarized by Konstantinidis et al. report that with full feedback the hot-stove effect is “no longer relevant” in their task, because participants can watch the risky option’s outcomes even when they do not choose it. (Konstantinidis et al., 2022) (Konstantinidis et al._1, 2022) Cao et al. make the same point more generally: when counterfactual feedback eliminated sampling asymmetry, the decision bias they studied vanished. (Cao et al., 2023)

So, on the narrow question of **preventing self-reinforcing sampling mistakes**, the answer is mostly yes: richer feedback helps because it breaks the link between avoidance and ignorance. But that does **not** mean full feedback always improves performance. Weiss-Cohen et al. found that removing the hot-stove effect with full feedback made people take more risk and earn lower bonuses in their task, because the foregone outcomes weakened the warning effect of rare bad outcomes. (Weiss-Cohen et al., 2020) In other words, full feedback is good at stopping one kind of error, but it can also remove a conservative bias that was sometimes protecting people from losses.

The broader learning literature points in the same direction. In theory, bandit feedback can still support good learning, but it is informationally weaker, and there are settings where it no longer matches what richer feedback can achieve. Lykouris et al. note that full-feedback models are substantially easier than bandit models, and Cheng et al. report that bandit feedback can fail to achieve optimal regret when available resources are large enough that extra information should matter. (Lykouris et al., 2019) (Cheng et al., 2023) The practical synthesis is therefore: **bandit feedback is the regime in which hot-stove dynamics naturally arise and persist; full or counterfactual feedback is the main tool for breaking them, because it supplies the missing evidence about unchosen options.**

Evidence

(Lykouris et al., 2019)

Corruption Robust Exploration in Episodic Reinforcement Learning

> *"We focus on the bandit feedback setting, where after each step the learner only observes the outcomes (reward and the new state) of the chosen actions. This is in contrast to the generative setting, where the learner has access to a simulator that allows her to sample transitions and rewards from any state-action pairs she chooses"*
>
> *"Related is the full-feedback setting, where after each step the learner observes the rewards and transition probabilities for all state-action pairs, not only for the current state and the chosen action. This feedback model is known to be substantially easier than bandit feedback, both in terms of techniques and in terms of the resulting regret bounds."*

(Weiss-Cohen et al., 2020)

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> *"Another behavioral phenomenon which can interfere in the learning process is the hot stove effect (Denrell et al., 2001)). 2 The hot stove effect states that bad outcomes reduce the attractiveness of an option, causing those options to be avoided (Biele et al., 2009). By avoiding an option after a bad outcome has occurred, the ability to collect more information about that option is severely constrained, as feedback from that option will no longer be observed. With no new information beeing collected, the bad outcome observed will remain the most recent (and most salient) observation associated with that alternative, decreasing its perceived value"*
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> *". The hot stove effect severely constrains the learning process by reducing the likelihood of selections from undesirable options, after a sequence of bad outcomes; in partial-feedback environments, there are no opportunities for further learning from unselected options, therefore eliminating learning about how unnattractive these options really are (i.e., how often the stove is hot; (Denrell et al., 2001). This process is likely to be related to the overweighting of rare events observed in decisions from descriptions, which would dictate that even though the described bad outcomes only occur rarely, individuals perceive them as happening much more often (4635002). In contrast, in paradigms, the hot stove effect is not sustained because individuals are able to learn from the foregone outcomes and therefore realize that the stove is not necessarily "hot" all the time. Eliminating the hot stove effect via full-feedback in our experiment was detrimental to performance, leading to more risk-taking, lower bonuses, and reduced impact of the descriptions, which acted as warning messages."*

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(Konstantinidis et al., 2022)

Patterns of choice adaptation in dynamic risky environments

"The effect suggests that early negative experiences have a prolonged effect on choice, leading to fewer selections from the option that generated these negative experiences (or outcomes). In the case of the increasing condition with partial feedback, participants would not experience the change in the relative value of the risky option (i.e., the "hot-stove", as it mostly returns 0 at the initial stages of the task) because they would under-explore this option in fear of receiving unfavourable 0 outcomes. However, with full feedback, the hot-stove effect is no longer relevant as participants can observe the outcomes of the "hot-stove" without experiencing them: such (foregone) outcomes are not consequential, but provide full information about the outcomes of a choice option."

(Konstantinidis et al._1, 2022)

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(Cao et al., 2023)

Decision bias and sampling asymmetry in reward-guided learning

"This, in turn, ameliorates the impact of sampling-based learning asymmetry, such as the over-exploitation of high-value options that hinders effective learning of other available options. Thus, these data provide compelling evidence that sampling-based asymmetries during learning are a central mechanism driving observed decision biases, as opposed to divisive normalization or other competing explanations"

". We found that the influence of target magnitude on choice accuracy effectively vanished under full-feedback conditions, $t(45) = .87$, $p > .1$"

". Importantly, when counterfactual feedback eliminated sampling asymmetry, the observed decision bias vanished."

(Cheng et al., 2023)

Understanding the Role of Feedback in Online Learning with Switching Costs

"One interesting finding is that while bandit feedback can still guarantee optimal regret when the budget is relatively limited, it no longer suffices to achieve optimal regret when the budget is relatively large."